

Digital Product Passports for circular manufacturing value chains, leveraging interoperable data ecosystems to meet EU ESPR compliance and enhance traceability.

Manufacturing × Product traceability and digital product passport × Blockchain and DLT · OS003 v3 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
69 is the world moving	36 can we play, can we win	LOW 1 named in the evidence	€5.0bn partial evidence · per year	NEXT regulation adopted or propose...	L3 23 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

Digital Product Passports (DPPs) are becoming mandatory under the EU's Ecodesign for Sustainable Products Regulation (ESPR), creating an urgent need for manufacturers to implement interoperable data ecosystems for lifecycle traceability. This opportunity targets manufacturing value chains—especially automotive, electronics, and plastics—that must achieve ESPR compliance while enhancing circularity. The time is now because technical standards are in place and the regulatory deadline is approaching.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€7.0bn €885m – €45.6bn	€5.0bn €635m – €32.7bn	€6.3m €794k – €40.9m	partial
Observed: tendered contract value, annualised	€20.9m €20.9m – €56.6m	€20.9m €20.9m – €56.6m	€26k €26k – €71k	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Manufacturing (EU27_2020)	238,122 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting Blockchain and DLT	52.7 % of enterprises (10+ employees)	proxy	Eurostat · isoc_cicce_usen2 · E_CC	2025
Annual value of one engagement	€337k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Size-weighted buyer base	39,220 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	0.1 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. Cloud computing services by NACE Rev. 2 activity series E_CC is used as a proxy for Blockchain and DLT: No DLT series at NACE level. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders.

Observed: tendered contract value, annualised

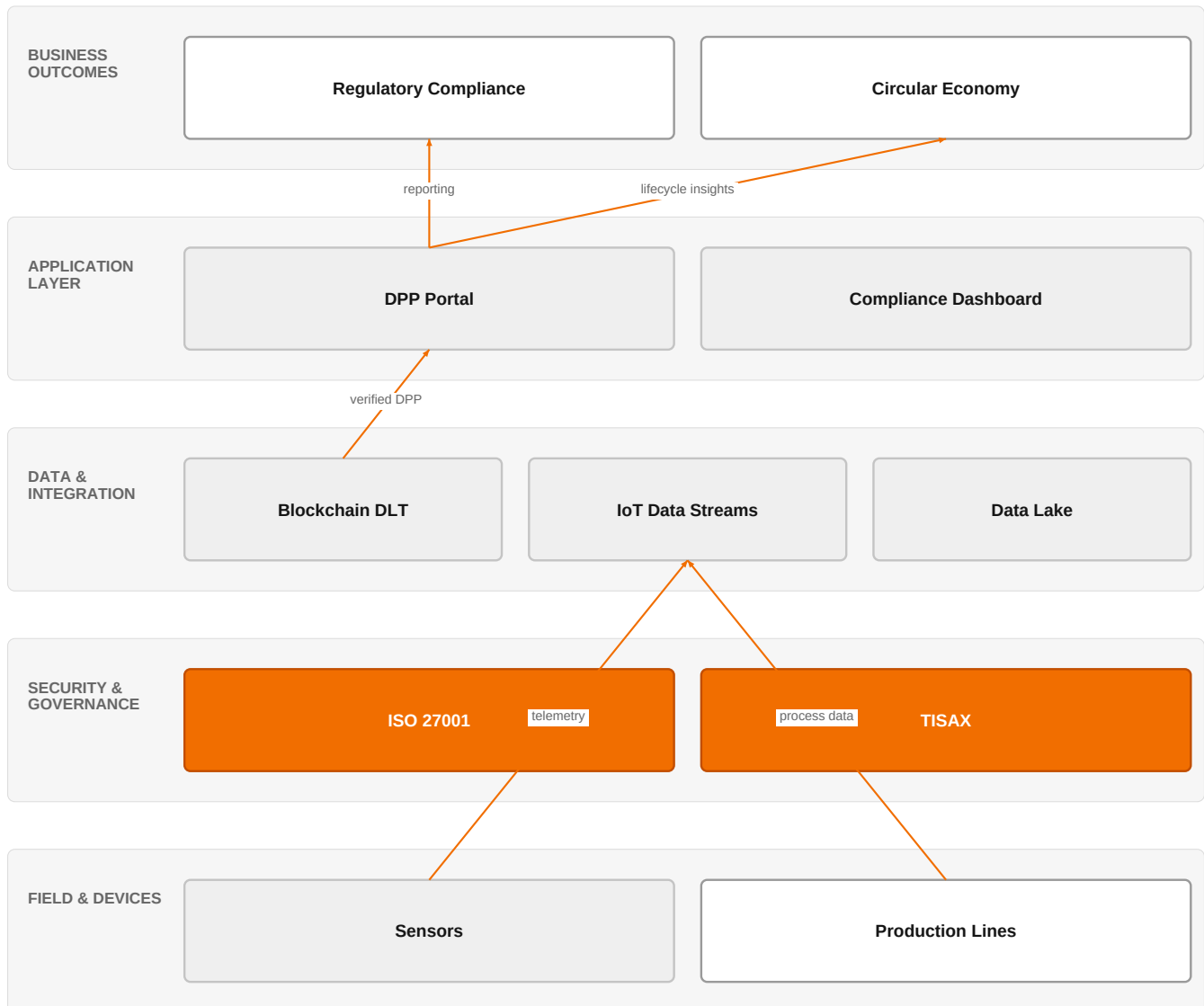
Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€20.9m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Assumed obtainable share of the serviceable market	0.1 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 14 matching notices, matched on vertical via the CPV crosswalk — \$4.5.2's warning applies: a crosswalk error lands directly

in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size.

The solution, in one picture

Digital Product Passport Solution Architecture



■ Orange asset ■ Partner ■ Customer-owned ■ Third party / to be sourced

This architecture shows how Orange's blockchain and IoT assets, secured by ISO 27001 and TISAX, enable a compliant and circular DPP ecosystem.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

The EU's Ecodesign for Sustainable Products Regulation (ESPR) mandates Digital Product Passports for a wide range of product categories, creating a regulatory driver for adoption. European technical standards for DPPs are now in place, and the EU's DPP registry raises new data demands. However, full integration is hampered by organizational hurdles, fragmented standards, and slow adoption of digital tools, particularly in the automotive sector.

Sources: SIG-AA85D6CD0C67 openalex.org 2026-02-27; SIG-AF45FABC720F RECYCLING magazin 2026-08-03; SIG-A3D483674C44 Packaging Insights 2026-08-03; SIG-E4163170FA87 Systems 2026-06-17; SIG-FDD8BAC05BAA cordis.europa.eu 2026-07-01

Who buys, and why now

The primary buyer is the Chief Sustainability Officer or Head of Compliance in manufacturing firms, who feels the pain of impending regulatory deadlines and the complexity of data collection across fragmented supply chains. The trigger is the approaching ESPR compliance date, which forces them to act now. Operational pain includes managing lifecycle data from multiple suppliers, ensuring data quality, and integrating DPPs with existing ERP and IoT systems.

Sources: SIG-AA85D6CD0C67 openalex.org 2026-02-27; SIG-E4163170FA87 Systems 2026-06-17

Why it is hot now — every claim bound to a dated source

- Digital Product Passports are emerging as core infrastructure for sustainable manufacturing, enabling lifecycle transparency and circular economy practices. [SIG-36FFB2435F8A]
- Blockchain-based Digital Product Passports are becoming more widely used, with potential to mitigate consumer stress in luxury markets. [SIG-AC6FFC29A702]
- Blockchain-based Digital Product Passports are being studied for their role in mitigating consumer stress in online luxury markets. [SIG-AC6FFC29A702]
- Digital Product Passports are explored for enhancing transparency and supporting circular economy practices in the fashion industry. [SIG-FE1FBF23480C]
- Digital Product Passports are becoming core infrastructure for sustainable manufacturing, enabling lifecycle transparency and regulatory compliance. [SIG-36FFB2435F8A]
- Blockchain-based Digital Product Passports are becoming more widely used in various industries. [SIG-AC6FFC29A702]
- The study explores the role of the Digital Product Passport in enhancing transparency, promoting sustainable consumption, and supporting circular economy practices in the fashion industry. [SIG-FE1FBF23480C]
- Digital Product Passports are becoming mandatory in the EU for a wide range of product categories under the Ecodesign for Sustainable Products Regulation. [SIG-AA85D6CD0C67]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a DPP solution leveraging its Blockchain and DLT technology, IoT experts, and Orange Group researchers. The engagement would start with a compliance assessment, then design a DPP data model aligned with ESPR and Gaia-X frameworks, integrate IoT data streams for real-time traceability, and deploy a blockchain-based DPP registry. Orange's ISO 27001 and TISAX certifications ensure security and automotive-specific compliance. References like Saint-Gobain Glass and Colgate-Palmolive demonstrate manufacturing credibility.

Why Orange can win

Orange can win by combining its Blockchain and DLT technology with IoT expertise and strong security certifications (ISO 27001, TISAX). Its experience with manufacturing clients like Saint-Gobain Glass and Colgate-Palmolive provides proof points. However, the assets are thin—no specific DPP platform or vertical solution is named, so Orange must position as an integrator of its own components plus partner technologies.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Blockchain and DLT	technology	L3	no offer path, but adjacent assets or Orange research exist	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
TISAX	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Colgate-Palmolive	reference	SUP	published reference in this vertical, different use case	unconfirmed
Mondelez International	reference	SUP	published reference in this vertical, different use case	unconfirmed
Saint-Gobain Glass	reference	SUP	published reference in this vertical for this use case	unconfirmed
Skytale	reference	SUP	published reference in this vertical, different use case	unconfirmed

6 further published reference(s) in this vertical are linked to this topic and are listed in the radar.

Competitive landscape

LOW competitive intensity (score 2.9; 1 of 8 listed players appear in this topic's own evidence)

One or two adjacent players. Orange would be shaping the space rather than fighting for it.

Competitors include Databricks, which is named in the evidence for its data platform capabilities; Accenture and Capgemini as global systems integrators with deep manufacturing expertise; Deutsche Telekom/T-Systems and Vodafone as telecom peers; Bosch and Honeywell as industrial OT specialists; and Dassault Systemes/3DS Outscale as a sovereign cloud provider. Each brings strengths in data, integration, or industrial domain knowledge, and the customer will compare Orange against these.

Sources: SIG-BC952EFA76FC Databricks 2026-07-27

Competitor	Category	Position here	Basis	Evidence
Databricks	Data and AI platform	named in this topic's evidence	evidenced	SIG-BC952EFA76FC
Accenture	Global systems integrator	active in Manufacturing; competes in OX: Smart Industries	structural	—
Capgemini	Global systems integrator	active in Manufacturing; competes in OX: Smart Industries	structural	—
Deutsche Telekom / T-Systems	Telecom operator peer	active in Manufacturing; competes in OX: Smart Industries	structural	—
Vodafone Business	Telecom operator peer	active in Manufacturing; competes in OX: Smart Industries	structural	—
Bosch	Industrial / OT specialist	active in Manufacturing; competes in OX: Smart Industries	structural	—
Dassault Systemes / 3DS Outscale	Sovereign cloud provider	active in Manufacturing; competes in OX: Smart Industries	structural	—
Honeywell	Industrial / OT specialist	active in Manufacturing; competes in OX: Smart Industries	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 Which product categories in your portfolio are in scope for ESPR, and what is your target compliance date?
- 2 How do you currently collect and manage lifecycle data from your suppliers—is it manual, fragmented, or centralized?
- 3 Have you started any DPP pilot or proof-of-concept? If so, what were the main technical or organizational hurdles?
- 4 What is your current level of IoT adoption in manufacturing—do you have real-time data streams from production lines?
- 5 Who in your organization owns the DPP initiative—compliance, IT, or sustainability?
- 6 What is your expectation for DPP beyond compliance—do you see it as a strategic asset for circularity or just a regulatory burden?

Objections you will meet

They will say	You can answer
We already have a data platform from Databricks or similar; why do we need Orange?	Databricks provides strong data analytics, but DPP requires end-to-end traceability across supply chains, including IoT integration and blockchain-based provenance. Orange brings the connectivity, IoT expertise, and security certifications to build a complete DPP ecosystem, not just a data lake.
DPP standards are still evolving; we are waiting for clarity.	European technical standards are now in place, and the EU registry is raising data demands. Waiting risks non-compliance. Orange can help you build a flexible DPP architecture that adapts to final standards, using Gaia-X frameworks for interoperability.
We have internal IT teams that can handle this.	DPP implementation involves complex data sharing across organizational boundaries, requiring governance and security. Orange's ISO 27001 and TISAX certifications, plus experience with manufacturing clients, reduce risk and accelerate time-to-compliance.

Risks and unknowns

The main risk is that DPP standards are still evolving, and interoperability across supply chains remains a challenge. Organizational hurdles and fragmented standards could stall adoption. Unknowns include the final technical specifications for DPPs and how they will integrate with existing systems. Orange's ability to deliver a complete solution depends on partner ecosystems that are not yet fully defined.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Develop a bid brief for a European automotive manufacturer, outlining a Digital Product Passport solution based on our Blockchain and DLT technology, referencing the EcoPlast project's approach to circularity and our ISO 27001 and TISAX certifications to address security and compliance.
Sales	In a conversation with a European automotive manufacturer's sustainability or digital transformation lead, mention our experience with Saint-Gobain Glass and Colgate-Palmolive in traceability and circularity, and ask about their ESPR readiness and any pilots they are considering.
Strategist / Innovator	Deep-dive into the EU ESPR timeline and the specific data requirements for Digital Product Passports in the automotive sector, leveraging Orange Group researchers and IoT experts to map our current capabilities against the compliance needs.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-FDD8BAC05BAA	2026-07-01	cordis.europa.eu	1	Driving the Circular Transformation of the European Automotive Industry with Skills, Digital To...
SIG-E4163170FA87	2026-06-17	Systems	1	Unlocking Digital Product Passport Integration: Multidimensional Hurdles in Supply Chains
SIG-63476A345BB2	2026-06-01	cordis.europa.eu	1	Fostering Resilient and Auto-Compliant Manufacturing Value Chains through Advanced Digital Prod...
SIG-AC6FFC29A702	2026-06-01	openalex.org	1	Positive Technology for Consumer WellBeing: The Role of BlockchainBased Digital Product Passp...
SIG-9B4563021774	2026-04-21	openalex.org	1	Unlocking Sustainable Supply Chains Through Blockchain Traceability: The Strategic Roles of Tra...
SIG-36FFB2435F8A	2026-04-20	openalex.org	1	A Comprehensive Review of Digital Product Passports in Sustainable Manufacturing: Current State...
SIG-AA85D6CD0C67	2026-02-27	openalex.org	1	Digital product passports
SIG-F47C31AB60CB	2026-02-14	MDPI AG	1	Dynamic Traceability and Reliability Management: Integrating the Digital Product Passport with ...
SIG-BA196FF05244	2026-02-03	Springer Science and Busi...	1	Digital Product Passport for Circular Economy: A Review of Semantic Web Vocabularies and Requir...
SIG-FE1FBF23480C	2026-02-01	openalex.org	1	Digital product passports for cleaner production: Economic evidence from producers and consumer...
SIG-4B5C641F1D4C	2026-01-01	Elsevier BV	1	Digital Product Passport and Internet of Things Integration in Supply Chain Management
SIG-CC28A63D0BB7	2026-01-01	openalex.org	1	From Compliance to Circularity: Stakeholder Perspectives on Adopting Digital Product Passports ...
SIG-EC1CEFFD337A	2025-10-15	openalex.org	1	Digital Product Passport in support of Data-Driven End-of-Use Strategies—a systems design persp...
SIG-40DC35D6534F	2025-10-14	openalex.org	1	Predictive Generation of Digital Product Passport as Decision Support in Circular Networks
SIG-A61B4DE2FFB7	2025-09-19	openalex.org	1	Role of Certifications and Labelling in Ensuring Authenticity and Sustainability of Fermented M...
SIG-75DE7DFC9EA4	2025-06-01	cordis.europa.eu	1	Empowering Circular Operations in the Automotive Plastics Value Chain
SIG-A9A7A4BCDCC6	2025-05-01	cordis.europa.eu	1	Demonstrating circular value chains for complex plastics via collection, reuse, repair, remanuf...
SIG-A3D483674C44	2026-08-03	Packaging Insights	2	EU's Digital Product Passport Registry raises new data demands - Packaging Insights
SIG-AF45FABC720F	2026-08-03	RECYCLING magazin	2	Digital Product Passport: European Technical Standards Now in Place - RECYCLING magazin
SIG-BC952EFA76FC	2026-07-27	Databricks	2	The EU Digital Product Passport: a traceability deadline - Databricks
SIG-C4998B3557D1	2026-07-22	INSIGHT EU MONITORING	2	EU Commission launches Digital Product Passport Registry - INSIGHT EU MONITORING
SIG-4FE0689B82DF	2026-04-30	Packaging Digest	2	EU Digital Product Passport Impact on US Supply Chains - Packaging Digest
SIG-AB0CFE062092	2026-04-27	Packaging Gateway	2	EU packaging sector prepares for digital product passport rollout - Packaging Gateway

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: diagram (2 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
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Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:15+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:18+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-18T20:40:27+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.